

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	01 October 2026
Team ID	SWTID-2026-2085
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

The diagram above follows the guidelines in your template. The client runs locally in the browser, the server and database are cloud-hosted, and the data-storage components are shown separately. The flow is: the user works in the React UI, requests pass the auth middleware, the upload parser stages the spreadsheet rows, and the transform engine maps and validates them into the target records.

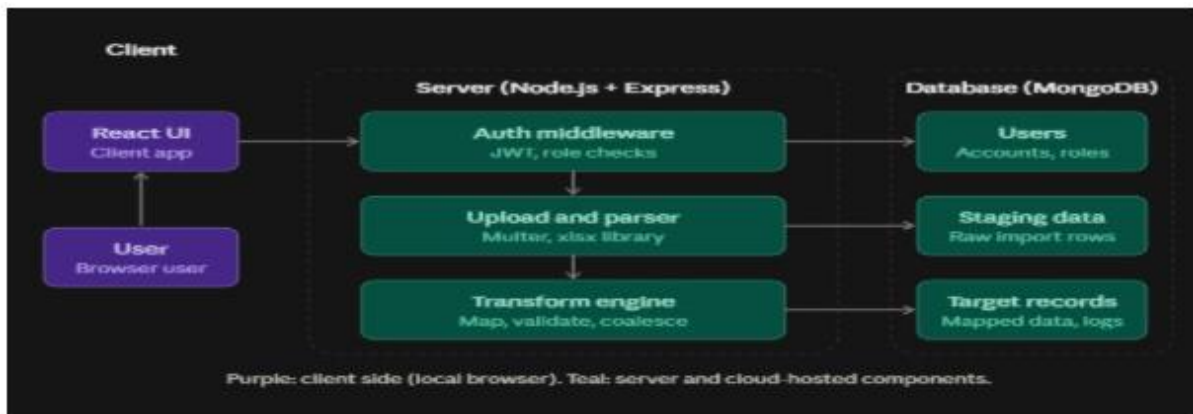


Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web UI for login, file upload, data preview, transform map editor, and import history	React.js, HTML, CSS, JavaScript
2	Application Logic-1	Authentication and authorization (JWT, role-based access)	Node.js, Express.js, bcrypt, jsonwebtoken
3	Application Logic-2	Spreadsheet upload and parsing (.xlsx / .csv) into staging	Multer, <code>xlsx</code> library
4	Application Logic-3	Transform engine: field mapping, validation, coalesce, insert/update	Node.js (JavaScript)
5	Database	Stores users, transform maps, and import history	MongoDB, Mongoose

6	Cloud Database	Hosted database service	MongoDB Atlas
7	File Storage	Temporary storage of uploaded spreadsheet files	Local file system (server <code>uploads/</code> folder) or cloud storage such as AWS S3
8	External API-1	Email/OTP confirmation and Gmail sign-in	Nodemailer or an OTP service, Google OAuth
9	External API-2	Not used [replace if you add one]	N/A
10	Machine Learning Model	Not applicable; the project uses rule-based mapping, not ML	N/A
11	Infrastructure (Local / Cloud)	Frontend and backend hosted on the cloud, with local development and testing	Render / Vercel / AWS, Git & GitHub, Postman

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	React, Express, Node.js, Mongoose, Multer, <code>xlsx</code>	MERN stack
2	Security Implementations	Password hashing, JWT tokens, role-based access control, file type and size validation, environment variables for secrets, HTTPS	bcrypt, JWT, CORS, Helmet, dotenv
3	Scalable Architecture	Three-tier architecture (client, server, database) where each tier scales independently; background job queue for large imports	REST API, MongoDB indexing, [optional: Bull/Redis]
4	Availability	Cloud hosting with database backups for recovery	MongoDB Atlas, cloud hosting
5	Performance	Batch inserts (<code>bulkWrite</code>), indexed coalesce fields, and streaming for large files	Mongoose bulk operations, pagination